TUNIS BUSINESS SCHOOL
UNIVERSITY OF TUNIS

IT-360 Information Assurance and Security

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Cyber Security Project

3rd Deliverable

North African Hotel Pricing and Satisfaction Monitoring using Web Scraping

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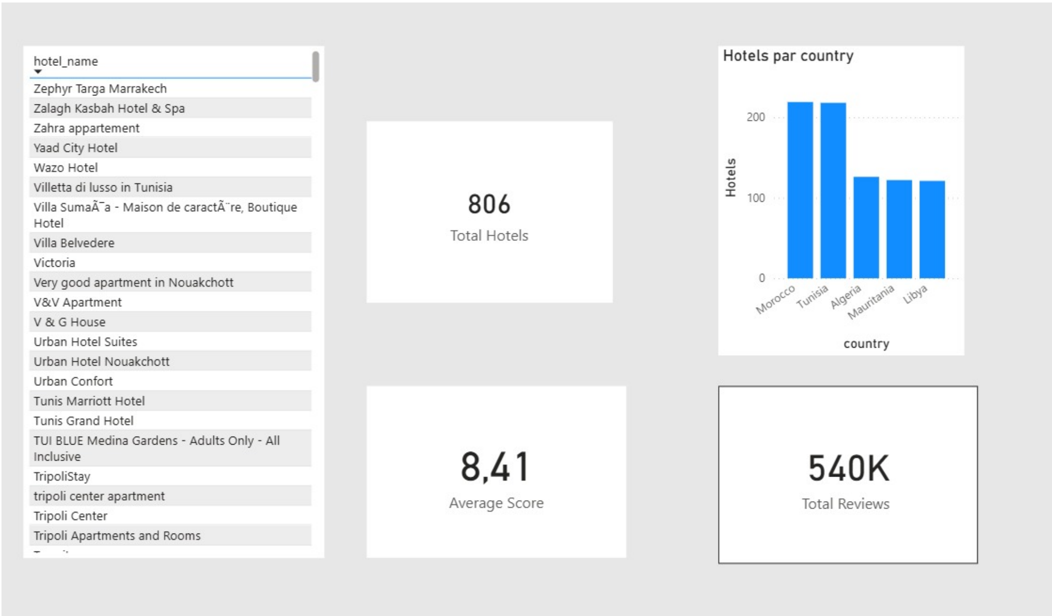
1. Introduction

This report constitutes the third and final phase of a cybersecurity-oriented web scraping project focused on the North African hotel industry. Having completed website scraping and data cleaning in previous deliverables, this stage concentrates on **data visualization** using **Microsoft Power BI**.

The interactive dashboards developed in this phase transform raw scraped data into actionable business intelligence, covering four analytical dimensions: a regional overview, price distribution patterns, the spatial relationship between hotel location and price, and review quality metrics. Together, these dashboards provide a comprehensive picture of pricing dynamics and guest satisfaction across the North African hospitality market.

2. Dashboard 1 — Regional Overview



2.1 Dashboard Description

The Overview dashboard provides a high-level summary of the entire dataset through three KPI cards and an interactive hotel slicer, establishing the foundational context for all subsequent analyses.

2.2 Interpretation of Visuals

2.2.1 Total Hotels (KPI Card)

The dataset covers **806 hotels** in total across North Africa, providing a substantial and representative sample for regional analysis.

2.2.2 Average Score (KPI Card)

The portfolio-wide average score sits at **8.41 out of 10**, confirming that the overall quality of hotels across the region is consistently strong and falls within the *Excellent* tier.

2.2.3 Total Reviews (KPI Card)

A combined **540,000 reviews** underpin these scores, giving the ratings strong statistical credibility — this is not a small or biased sample.

2.2.4 Hotel Name Slicer (Left Panel)

The interactive slicer lists all 806 hotels alphabetically, allowing any single property to be isolated across all visuals. The variety of names visible — from large branded chains such as *Tunis Marriott* and *TUI BLUE* to small apartments and boutique stays — confirms that the dataset captures the full accommodation spectrum, from budget to luxury.

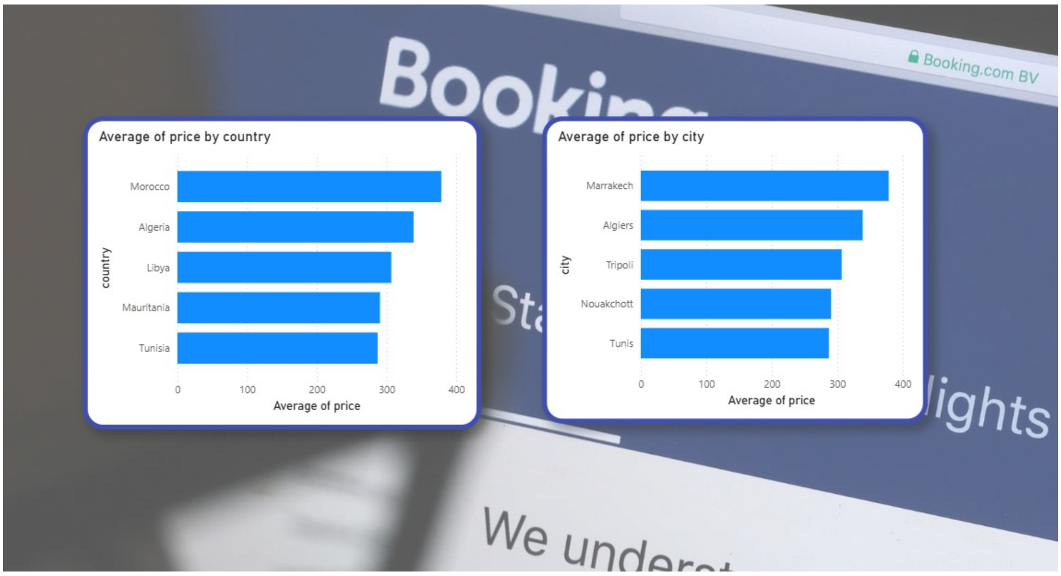
Key Insight

With 806 hotels and over half a million verified reviews, the dataset is statistically robust. The 8.41 average score confirms that North African hospitality broadly meets international quality benchmarks, yet the diversity of property types demands segmented analysis to avoid misleading aggregate conclusions.

Business Recommendation

Hospitality authorities and tourism boards should use this broad, high-quality dataset as a baseline for regional benchmarking. Properties that fall significantly below the 8.41 regional average should be targeted for improvement programmes or promotional audits. Investors considering new hotel developments can treat this average as the minimum quality bar required to remain competitive in the regional market.

3. Dashboard 2 — Price Distribution



3.1 Dashboard Description

The Price Distribution dashboard examines how average nightly rates vary across countries, cities, individual hotel brands, and seasons, revealing both market-level disparities and within-market pricing strategies.

3.2 Interpretation of Visuals

3.2.1 Average Price by Country

Morocco leads all countries with the highest average price (~370 TND), while Tunisia is consistently the most affordable despite being a major tourism hub. Algeria, Libya, and Mauritania sit in a tight mid-range band, suggesting relatively uniform pricing across those three markets.

3.2.2 Average Price by City

Marrakech is the most expensive city (~370 TND), consistent with Morocco's country-level lead. Tunis sits at the bottom (~295 TND), reinforcing Tunisia's budget-friendly positioning across the dataset.

3.2.3 Average Price by Hotel Name

A clear two-tier market exists. *TUI BLUE Medina Gardens* and *Iberostar Waves Club Palmeraie* sit between 800–870 TND — likely all-inclusive resorts — while most other named hotels fall well under 400 TND. Country-level averages are being pulled upward by a small number of premium outliers.

3.2.4 Average Price by Season

May (Spring) is the most expensive season (~340 TND), yet the gap between all three seasons is only about 40 TND. July, despite being peak summer, is actually the cheapest — indicating that hotels across the region apply very little dynamic seasonal pricing, which represents a missed revenue opportunity.

Key Insight

The market is structurally bifurcated: a thin layer of all-inclusive luxury resorts commands rates two to three times the regional average, while the majority of properties compete in a narrow, undifferentiated mid-to-budget tier. Crucially, the near-flat seasonal pricing curve reveals that most hotels are not capitalising on demand peaks.

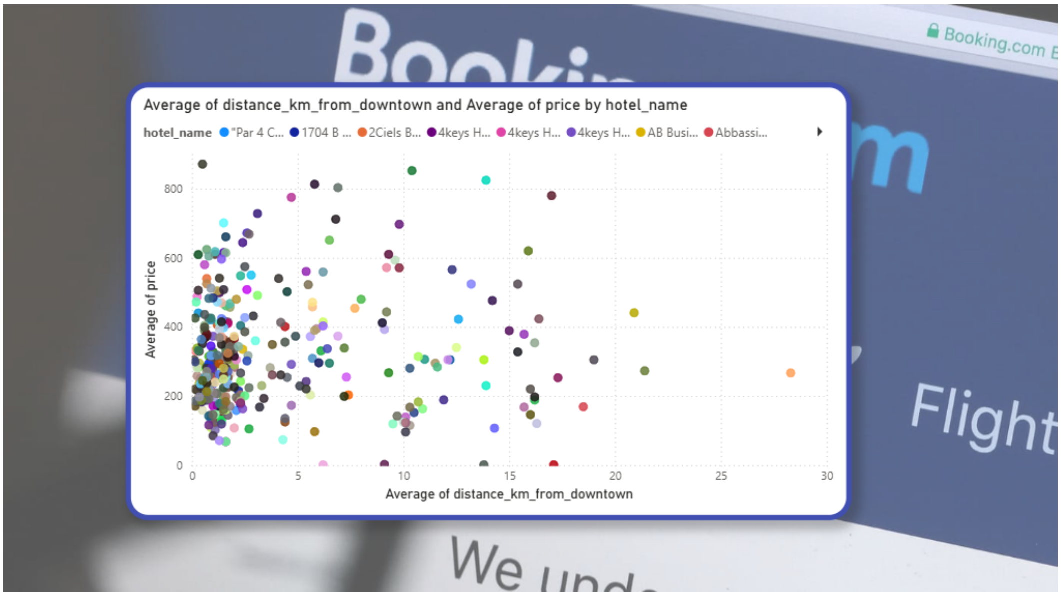
Business Recommendation

For hotel operators: Implement demand-based dynamic pricing for peak periods, particularly July and August. A 15–20% rate increase during confirmed demand peaks would generate material incremental revenue without significant occupancy risk, given the strong regional review scores.

For policymakers and tourism boards: Tunisia's pricing gap relative to Morocco represents both a competitive advantage and a risk. Positioning Tunisia as a value destination should be deliberate and supported by quality assurance — not simply a default outcome of structural underpricing.

For investors: The dominance of the luxury tier by only two properties in Morocco signals an undersupply of premium offerings in other North African countries, representing a potential investment gap.

4. Dashboard 3 — Distance from Downtown vs. Price



4.1 Dashboard Description

This dashboard investigates whether physical proximity to a city centre is a reliable predictor of hotel pricing, using a scatter-plot of all 806 properties mapped by distance and rate.

4.2 Interpretation of Visuals

4.2.1 Overall Pattern

The vast majority of hotels cluster within 0–5 km of downtown, and within that zone prices range from near zero to over 700 TND. Being centrally located says very little about what a hotel charges.

4.2.2 No Clear Distance–Price Correlation

Expensive hotels appear at nearly every distance band, including 15–30 km from downtown. These high-priced outliers far from city centres are almost certainly beachfront or resort properties, where amenities drive pricing more than urban proximity.

4.2.3 Key Takeaway

Budget hotels exist exclusively near city centres, while luxury pricing appears at all distances. Distance from downtown alone is not a reliable price predictor — the market needs to be segmented into *city hotels* versus *resort hotels* before any meaningful pricing analysis can be conducted.

Key Insight

The absence of a distance–price relationship confirms that the North African hotel market operates along two fundamentally different value propositions: urban convenience versus resort amenities. Treating the market as a single segment produces misleading conclusions in all downstream analyses.

Business Recommendation

For data analysts and future research: All subsequent pricing models should incorporate a binary `hotel_type` feature (city vs. resort) before any regression or clustering is applied. Failure to do so will depress model accuracy and obscure genuine pricing signals.

For booking platforms: Filtering and search logic should expose this distinction to users. Travellers searching by proximity to downtown likely have different price expectations than those searching by beach or resort, and the user experience should reflect that difference.

For hotel operators in suburban/coastal locations: The data confirms that distance from downtown is not a liability — amenity quality and category positioning matter far more. Marketing budgets should be directed at communicating unique resort value rather than competing on location.

5. Dashboard 4 — Review Quality, Volume & Score Analysis

5.1 Dashboard Description

The fourth dashboard consolidates guest satisfaction data through multiple lenses: KPI cards, seasonal score trends, review label distribution, country-level score benchmarking, and a review count versus score scatter plot.

5.2 Interpretation of Visuals

5.2.1 KPI Cards

The portfolio averages a strong **8.41 out of 10** across 806 total hotels, placing it firmly in the *Excellent* tier and signalling broad guest satisfaction across the region.

5.2.2 Average Score by Season (Line Chart)

May (Spring) scores highest at **8.45**, July dips slightly, and September (Low Season) records the lowest scores. Spring is the only season where both price and guest satisfaction peak simultaneously — making it the most strategically valuable period in the entire dataset.

5.2.3 Review Distribution by Label (Pie Chart)

Excellent (36.6%) and *Wonderful* (33.1%) together account for nearly 70% of all reviews — a healthy, quality-skewed distribution. The *Exceptional* tier is barely a sliver, meaning no hotel has yet broken through to consistently perfect guest experiences.

5.2.4 Average Score by Country (Bar Chart)

Libya leads at ~8.6 and Morocco follows at ~8.5, both consistent with their premium market positioning. Algeria is the outlier — it is the second most expensive country yet scores the lowest at ~8.2, signalling a value-for-money gap that risks guest dissatisfaction.

5.2.5 Review Count vs. Score by Hotel (Scatter Plot)

One hotel stands sharply apart with ~28,000 reviews and a score just above 5. This high-volume, low-quality outlier is likely a large budget property and is pulling down regional averages disproportionately.

Key Insight

The convergence of peak pricing and peak satisfaction in Spring (May) is the single most actionable data point in the entire dataset. Algeria's value-for-money gap and the single low-scoring high-volume outlier are the two most significant quality risks currently distorting regional averages and investor perception.

Business Recommendation

For hotel operators — Spring strategy: May represents the optimal window for premium package launches, loyalty programme promotions, and yield maximisation, as guests are both more willing to pay and more satisfied. Revenue and marketing strategies should be aligned to this period above all others.

For the Algerian hospitality sector: The combination of above-average pricing and below-average scores represents an urgent value-for-money issue. Operators and tourism authorities should audit service standards at mid-to-high-price properties and consider whether rate reductions or quality investments are the more viable corrective path.

For platform integrity: The single property with 28,000 reviews and a score above 5 should be investigated for review authenticity. Such a disproportionate volume of reviews for a low-quality property may indicate review manipulation, which would compromise the integrity of regional benchmarks. Platforms should apply anomaly detection to flag outlier review volumes.

For quality benchmarking: The near-absence of *Exceptional* ratings across the region suggests that no hotel has yet established a clear quality leadership position. This represents a first-mover opportunity for any operator willing to invest systematically in service excellence.

6. Conclusion

This Power BI analysis of 806 North African hotels, backed by over 540,000 guest reviews, reveals a hospitality market with strong overall quality credentials but several structural inefficiencies that represent both risks and opportunities.

The key findings can be summarised as follows:

- The regional quality baseline is high (8.41/10), but aggregate scores mask important country-level and property-level disparities.
- Morocco commands the highest prices; Tunisia the lowest — yet satisfaction levels do not always align with price position, as Algeria's outlier status demonstrates.
- Seasonal pricing is remarkably flat across the region, representing a systematic missed revenue opportunity, particularly given the measurable quality peak in Spring.
- The market is structurally bifurcated into city hotels and resort hotels; analyses that ignore this distinction are methodologically unreliable.
- A small number of premium properties and one large low-scoring outlier exert disproportionate influence on regional averages and require isolated treatment in any further analysis.

These insights, derived entirely from publicly scraped and cleaned data, demonstrate the significant business intelligence value that can be extracted through ethical and methodical web scraping — reinforcing the relevance of such techniques within the information assurance and security discipline.